Vadim Atlassov

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Research Interests: Diffusion Models · Multimodal Learning · Medical Imaging · Foundation Models

PUBLICATIONS

V. Atlassov, I. Schlattner, Y. Omarov, H. Ju, M.-H. Lee. "Controllable Diffusion-Based Image Generation for Failure Diagnosis of Reinforced Concrete Beam–Column Joints." *Journal of Building Engineering*, 128 (2026) 116466. **Accepted**. IF: 8.1, Q1.

I. Schlattner, **V. Atlassov**, A. Abdrakhmanov, M.-H. Lee. "Latent-Aligned Scalable Interpolant Transformer for Controllable Chest X-ray Synthesis." *Scientific Reports, Nature Portfolio*. **Accepted**. (2026)

V. Atlassov, M.-H. Lee. "RECON: A Vision-Language Model for Automated Damage Assessment and Reporting of RC Beam–Column Joints." *In Preparation*. (2026)

M.-H. Lee, A. Abdrakhmanov, **V. Atlassov**, I. Schlattner, D. Kim, H. Ju. "A Benchmark Image Dataset of Reinforced Concrete Beam–Column Joint Failures." *Scientific Data, Nature Portfolio (Major Revision)*. (2026)

EDUCATION

M.Sc. in Electrical & Computer Engineering, Nazarbayev University 2023
GPA: 3.21/4.00

B.Eng. in Electrical and Computer Engineering, EKTU 2020
GPA: 3.71/4.00 (Magna Cum Laude)

Exchange Student, Brno University of Technology, Czech Republic 2019

RESEARCH EXPERIENCE

Research Assistant April 2025 – Present
Human-Computer Interaction Lab, Nazarbayev University
Supervisor: Prof. Min-Ho Lee

- **Diffusion-Based Generative Modeling**
  Developed and trained controllable diffusion model for structured image generation using spatial, geometric, textual, and mask conditioning.
- **Controllable Mammographic Image Synthesis**
  Developed controllable diffusion framework for mammography with structured representations of **breast contour, fibroglandular density, and focal lesions**; implemented multimodal anatomical conditioning and lesion-focused inpainting for controllable synthesis and data augmentation.
- **Latent-Aligned Chest Radiograph Generation**
  Designed a two-stage Latent-Aligned Diffusion model for controllable chest X-ray synthesis with structured conditioning and counterfactual pneumonia ↔ healthy generation.
  FID **45.38** · MS-SSIM **0.736** · Dice **0.6564**
- **Physics-/Domain-Conditioned Diffusion**
  Developed a geometry-conditioned diffusion framework using structural priors and a frozen Stable Diffusion backbone for controllable image generation.
  KID **0.0628** · SSIM **0.790** · Failure-type acc. **85.8%** · Rebar-exposure acc. **95.2%**
- **Vision-Language Foundation Models (RECON)**
  Developed a domain-specific VLM combining DINOv2, Q-Former, Mixture-of-Experts adaptation, and LLaMA-3B for automated visual understanding and report generation.
  BLEU-4 **0.3023** · CIDEr **1.0262** · ROUGE-L **0.4381**

Research Assistant September 2022 – June 2023
ALARIS Robotics Lab, Nazarbayev University
Supervisor: Prof. Almas Shintemirov
Co-supervisor: Prof. Berdakh Abibullaev

- Developed a CNN-based 6-DoF pose estimation system for UR5 manipulation in degraded visual environments; **10.2 mm MAE** in 3D trajectory alignment.

TECHNICAL SKILLS

Generative AI: Diffusion/Flow Matching Models, ControlNet, T2I-Adapter, VAE, VLMs/LLMs
Foundation Models: DINOv2/v3, CLIP, BLIP-2, Mixture-of-Experts, Self-Supervised Learning, PEFT/LoRA
Deep Learning: PyTorch, CUDA, CNNs, Transformers, Representation Learning
Systems & Development: Python, ONNX, Docker, Linux, Git, VIM

LANGUAGES

English — IELTS Academic: 8.0 (2026)
Russian — Native
Kazakh — B2
Czech — B2

HONORS & AWARDS

Graduate Academic Merit Award, Nazarbayev University.
2nd highest ranking in undergraduate graduating class, EKTU (2020).

PROFESSIONAL EXPERIENCE

Software Engineer, VEON Ltd (Beeline Kazakhstan) July 2023 – April 2025
Designed and deployed billing systems serving 11.6M+ users with 99.9% uptime.